Literature dissection

Spring 2026

Lukas Lescano, Garrett Mellinger, Minh Tri Ngo

2026-05-29

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Photo documentation

Individually:

1. Find the document titled "Data Description: Photo Documentation of the North Campus Open Space Restoration Project".
2. Read about the photo documentation process.
3. Find 1-3 photo points relevant to your study using the ArcGIS web map.

As a group:

4. Discuss which photo point to focus on based on its relevance to your study question or context.
5. Decide on a photo point.
6. Find 2-4 photos at that photo point to represent seasonal changes within a water year.
7. In the table below, fill in the information for each photo at the photo point.

Table

Photo point number	Date of photo	Season	Direction of photo	Link to photo

|(14)| April 2024 | Spring | W | Link to Photo #14 West |

|(14)| April 2024 | Spring | SW | Link to Photo #14 Southwest |

|(15)| April 2024 | Spring | W | Link to Photo #15 West |

|(15)| April 2024 | Spring | SW | Link to Photo #15 Southwest |

|(08)| April 2024 | Spring | E | Link to Photo #08 East |

|(08)| April 2024 | Spring | SE | Link to Photo #08 Southeast |

8. Describe in 5-7 sentences total:

- which water year you chose to examine
- how that water year is relevant to your study
- which photo point you chose to examine
- how that photo point is relevant to your study
- what changes you see in vegetation, animals, environmental characteristics from the photos you chose.

1. The water year we chose to examine was the water year 2024.
2. This water year is relevant to our study because in the scope of our focuses on the impacts of disturbance. We found that predisturbance was within the year 2023, and that post disturbance was within 2024 in the spring. Spring of 2024 is within the 2024 water year.
3. The photo point we chose was point 14.

4. This photo point is relevant to our study because it contains the most central location to native perennial grasslands. Each cardinal direction taken shows a majority of the grasslands. However, it's cardinal directions of West and Southwest have the most coverage of grassland transects of focus where pre and post disturbance. This allows for a better visual comparison of pre and post disturbance.
5. Changes I see in biotic and abiotic factors are primarily plant activity. Within the spring, the plants are green and alive while analyzing pre disturbance in October means many of the plant life is "dead" and yellow. Additionally, the treeline has drastically changed toward the boundary of the grassland, reducing as it goes from 2023 to 2024. Another change is the burn scares being visible within earlier water year photos while plant growth is more dominant in photos that have been taken in later parts. When comparing one year post disturbance, there are more grasses available in October of 2024 compared to October of 2023. There are also more shrub plants visible within a year of post disturbance compared to more grasses and low lying plants shown earlier in the water year. In terms of abiotic factors, there isn't much change as the oil facilities were removed over time, while other items like rocks remain in the exact same location. The fence is also in the same location and hasn't been adjusted.

Poster

Title: Evaluating Sheep Grazing as a Strategy for Restoring the North Campus Open Space (NCOS) Grasslands

Authors: Abby Nestor Faculty Advisors: Dr. Lisa Stratton and Dr. Carla D'Antonio

Year published: 2025

Permalink: Link to Poster

Dataset used (likely): self procured experimental datasets – greenhouse seed viability experiment, field biomass experiment

The dry biomass of native *Stipa pulchra* and *Deinandra fasciculata* and invasive species *Medicago polymorpha*, and *Melilotus indicus*, experienced a decrease following sheep grazing. The annual invasive grass *Festuca perrenis* increased its dry biomass following grazing. Additionally, T-test results suggest that the differences between the mean dry weights of the pre-grazing and post-grazing biomass samples of *F. perennis*, *M. polymorpha*, and *S. pulchra* are not statistically significant. Finally, The abundance of *M. polymorpha* individuals, very few *F. perennis* individuals, and complete lack of *S. pulchra*, *D. fasciculata*, and *M. indicus* individuals, suggests that the seeds of *M. polymorpha* are relatively durable and have high germination viability after passing through the sheep.

These findings show that intentional disturbance can affect plant communities in species specific ways, which connects to my question of how native and non native percent cover changes before and after disturbance at NCOS. The findings relate most closely to Figure 1 which compares native vs. non-native cover over time, and to Figure 2, which shows percent cover changes in key species after the 2023 cultural burn. The poster's findings suggest that disturbance does not produce one uniform response, which is important for evaluating whether disturbance is meeting restoration goals or favoring invasive cover.

Annotated bibliography

Background

You can find peer-reviewed articles by searching on Google Scholar, which is a large database of research papers, posters, and other materials.

Note: while it may be tempting to use AI to provide a list of papers for you, you should *not* be using AI for any component of this activity. You'll accomplish the objective of articulating the connection between your study and other study systems/questions by completing the difficult task of sifting through papers that may not be relevant to you before finding one that is.

Additionally, you can test out inserting citations in Quarto. See [this resource to insert a citation in a .qmd](#).

Directions

Individually:

1. Search Google Scholar for papers that might be relevant to your study. One recommendation for how to start: search for whatever your focal variables may be + "wetland restoration" (for example, "bird abundance wetland restoration", "water salinity wetland restoration", "aquatic invertebrate wetland restoration").
2. Find 4-5 papers that might be relevant based on title and what shows up in the search.
3. Read the abstracts of those 4-5 papers.
4. Decide which - if any - are actually relevant to your project. Make a note of these papers. Note that a paper may be relevant in that it shares a) taxonomic groups, b) geographic regions, c) research questions, d) research methodology - there are lots of ways that existing research can be tied into your project.
5. If step 4 did not yield any relevant papers, repeat steps 2-4.
6. Search for papers until you have 4 relevant papers per person (12 total for a 3 person group).

As a group:

7. Share the papers you found, discussing the main points of the paper and how they are relevant to your project.
8. Fill in the template for an annotated bibliography below.
9. In the "Main findings" or "Relevance to study" sections, insert a citation for the paper.

Paper 1

Title: Response of a Native Perennial Grass Stand to Disturbance in California's Coast Range Grassland

Authors:James W. Bartolome, Jeffrey S. Fehmi, Randall D. Jackson, Barbara Allen-Diaz

Main findings (2-4 sentences): Found no significant burning effects and grazing removal imparted a shift in plant community from more annual-dominated toward ore perennial-dominated vegetation. Individual perennial grass species responded differently according to genus and species. Under relatively mesic weather conditions it appears that grazing removal from Coast Range Grasslands with existing native perennial grass populations can increase their cover. However if N. pulchra is the sole existing population, spring season-restricted grazing should be equally effective at enhancing cover of the native grass species.

Relevance to study (2-4 sentences): This article found one method of disturbance created shifts in the plant community in a study location similar to NCOS. One of the methods we were looking at was grazing so the paper shows us guidance of possible outcome we could be looking at for results. Mention of burning having no impact also guides our expectation for results as well.

Paper 2

Title: Optimal prescribed burn frequency to manage foundation California perennial grass species and enhance native flora

Authors: Tina M. Carlsen, Erin K. Espeland, Lisa E. Paterson & Don H. MacQueen

Main findings (2-4 sentences): Annual burning maintained P. secunda number, resulted in significant expansion, the lowest thatch and exotic grass cover, the highest percentage of bare ground, but also the lowest native forb and highest exotic forb cover. Burning every 3 years maintained lower numbers of P. secunda plants, allowed for expansion, and resulted in the highest native forb cover with a low exotic grass cover while burning every 5 years and the control (burned once from a wildfire) resulted in a decline in P. secunda number, the highest exotic grass and thatch cover and the lowest percentage of bare ground. Perennial grass species management must also consider other native species life history and phenology to enhance native flora diversity.

Relevance to study (2-4 sentences): Paper shows that annual burning can result in expansion, low covers for other categories, and low native forb and high exotic forb which can also give us an aim on how our results should look. Burn intensity/frequency also gives us aim depending on how far and how often burn was in NCOS. Also gives us something for future consideration since perennial grass management also is needed for consideration.

Paper 3

Title:Prescribed Fire, Soil, and Plants: Burn Effects and Interactions in the Central Great Basin

Authors:Benjamin M. Rau, Jeanne C. Chambers, Robert R. Blank, Dale W. Johnson

Main findings (2-4 sentences): Burning resulted in increases in soil surface NH_4^+, NO_3^-, inorganic N, Ca^{2+}, Mn^{2+}, and Zn^{2+}. Increases in NO_3^-, inorganic N, and Zn^{2+} were also observed in deeper horizons. Burning did not affect aboveground plant biomass or nutrient concentrations in the first year with the exception of F. idahoensis, which had increased tissue P. By the second year, all species had statistically significant responses to burning. The most common response was for increased aboveground plant weight and tissue N concentrations. Plant response to burning appeared to be related to the burn treatment and the soil variables surface K^+, NO_3^-, and inorganic N.

Relevance to study (2-4 sentences): Shows other factors we could look at if available such as soil content. Also gives us aim for what data should look like as they mention how plant biomass looked like one year post disturbance. Also gives us insight into knowledge gap from other studies about what a 2nd year post disturbance looks like.

Paper 4

Title:Prescribed Fire Effects on Population Dynamics of an Annual Grassland

Authors:Sasha A. Berleman, Katharine N. Suding b, Danny L. Fry,James W. Bartolome, Scott L. Stephens

Main findings (2-4 sentences): 1) seed dispersal is an important factor in recovery dynamics and 2) wild oat fecundity significantly increases in the year after fire while medusahead and needlegrass fecundity seem minimally affected. Ultimately, managers should consider fire as a preferable first-entry tool and should thoroughly consider shape and size of planned burns, as well as what vegetation is present to play a role in post-treatment seed-dispersal dynamics.

Relevance to study (2-4 sentences): Articles give another factor (seed dispersal) to consider when looking at data or explanations that could explain such variances in data. Also shows how some species are affected differently from fires one year post disturbance. Also gives ideas for recommendations for future improvements toward NCOS.

Paper 5

Title:The effects of indigenous prescribed fire on riparian vegetation in central California

Authors:Hankins, D.L.

Main findings (2-4 sentences): Timing of burns matter. Fall burns support higher overall richness and native species richness; spring burns yield a decrease in overall richness, but an increase in native species richness; and summer burns contribute to an initial decline in overall and native species richness.

Relevance to study (2-4 sentences): Also gives reference to compare values received from NCOS; we can compare NCOS burns/disturbances with disturbances in article. Also used in same biome/region so results from article are more accurate to compare to.

Paper 6

Title: Prescribed Burn Impacts on California Coastal Prairie Seed Banks

Authors: Esther M. Galloway, Max R. Musto, Georgia L. Vasey, Karen D. Holl

Main findings (2-4 sentences): Pre- and post-burn soil samples from four Central Coast California prairie sites showed that non-native grass germinants significantly decreased after fire but remained high overall, suggesting single prescribed burns are insufficient to reduce invasive abundance. Non-native forbs showed a weaker post-fire decline while native forb and graminoid abundance did not change consistently. Burning during wet soil conditions was more effective at reducing non-natives than burning in dry conditions, though wet-season burns may adversely affect some natives.

Relevance to study (2-4 sentences): The Central Coast California coastal prairie setting is the strongest geographic match of any paper we identified, and the Fall 2024 sampling timeline overlaps with our post-burn monitoring period. The finding that single prescribed burns are insufficient to reduce non-native grass abundance contextualizes why we still observe high non-native cover post-burn in our data. Their seed bank focus complements our above-ground percent cover measurements as two approaches to the same management question.

Paper 7

Title: Grazing and Fire Management for Native Perennial Grass Restoration in California Grasslands

Authors: Menke, J.W.

Main findings (2-4 sentences): Time-controlled short-duration high-intensity sheep or cattle grazing in early spring removes alien annual plant seed and opens the sward canopy to allow light to reach perennial seedlings. Late spring prescribed burning reduces alien annual seed production and seedbank size while increasing perennial grass seedling establishment, though perennial grasses show reduced seed production the first year post-fire before recovering by year two or three. The author recommends burning every third or fourth year since alien annual species composition returns to pre-fire status within about three years.

Relevance to study (2-4 sentences): This paper directly addresses combined grazing and prescribed fire as restoration tools for native perennial grass restoration in California grasslands, making it the most directly relevant to our dual treatment design. Stipa pulchra is a focal species in this paper supporting our use of it as a native grass indicator, and the observation that perennials show reduced seed production the first year post-fire contextualizes our 2024 S. pulchra decline as an expected short-term response. The author also notes that competitive effects of annual clovers on native perennial seedlings are unknown, which is directly relevant to the clover increase we observe post-treatment.

Paper 8

Title: Evaluating Prescribed Fire Effect on Medusa Head and Other Invasive Plants in Coastal Prairie at Point Pinole

Authors: Bartolome, J.W., et al.

Main findings (2-4 sentences): A June 2016 prescribed burn at Point Pinole Regional Shoreline targeted medusa head and other invasive grasses across coastal prairie meadows. Pre- and post-burn vegetation monitoring used a nested frequency sampling approach along transects to document effects on target invasive grasses, other invasives, native perennial grasses, and common native forbs. The study documented effects of burning on coastal prairie species including Stipa pulchra, Brachypodium distachyon, and Distichlis spicata.

Relevance to study (2-4 sentences): The coastal prairie California grassland setting makes this study more ecologically comparable to NCOS mesa than most burn studies, and Stipa pulchra and Brachypodium distachyon appear in both datasets providing direct species-level comparability. The transect-based monitoring methodology is comparable to ours. We note the timing difference since their burn was conducted in June while ours was Fall 2023, which could affect which species are most impacted.

Paper 9

Title: North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024

Authors: Cheadle Center for Biodiversity and Ecological Restoration, UC Santa Barbara

Main findings (2-4 sentences): The report summarizes monitoring results from 2017 to 2024 across native and non-native plants, birds, small mammals, aquatic invertebrates, tree growth, and hydrology at NCOS. The project has restored over 40 acres of estuarine and palustrine wetlands and more than 60 acres of upland habitats including native grassland, coastal sage scrub, and vernal pools. The report documents site progression from barren excavated land in 2017 to a fully restored landscape by 2024.

Relevance to study (2-4 sentences): The report provided prescribed burn timing and treatment details essential to defining our pre and post burn comparison years. It also provided site context for the Mesa (GL) perennial grassland transects we analyzed. The 2017-2024 monitoring period offers a pre-treatment baseline that frames our 2021-2025 analysis window.

Paper 10

Title: Evaluating Sheep Grazing as a Strategy for Restoring the North Campus Open Space (NCOS) Grasslands

Authors: Cheadle Center for Biodiversity and Ecological Restoration, UC Santa Barbara

Main findings (2-4 sentences): Short-term rotational sheep grazing was used at NCOS to reduce thatch, invasive species seed banks, and invasive biomass to encourage native species growth. Pre-grazing and post-grazing biomass samples were compared and sheep pellet samples were assessed for seed viability over an eight-month greenhouse trial. The study evaluated the immediate impacts of grazing on key species abundance and the efficacy of the grazing regime in reducing invasive populations.

Relevance to study (2-4 sentences): This study is site-specific to NCOS making it directly applicable to our analysis and confirmed our selection of four focal species. We adapted their visualization approach to design our diverging bar chart showing percent point change in mean cover. We also adopted their paired t-test framework comparing pre- and post-treatment mean percent cover.

Paper 11

Title: Trajectories of vegetation-based indicators used to assess wetland restoration progress

Authors: Matthews, J.W., Spyreas, G.

Main findings (2-4 sentences): Restoration trajectories in 29 mitigation wetlands often deviated from the assumed monotonic increase toward reference conditions. Some floristic indicators rapidly reached or exceeded reference levels within five years, while others either failed to reach equivalency or took much longer than typical monitoring windows. Importantly, some indicators increased over the first 5 to 10 years and then declined, demonstrating that short monitoring windows can produce misleading assessments of restoration progress.

Relevance to study (2-4 sentences): The paper supports our interpretation that native cover declining post-burn does not necessarily indicate restoration failure, as non-monotonic trajectories are well-documented in restored ecosystems. It also justifies the need for longer-term monitoring beyond our current 2021-2025 window. We acknowledge that the wetland focus differs from our coastal grassland system, but the general principles about non-linear restoration trajectories still apply.

Paper 12

Title: A decade-long study of repeated prescription burning in California native grassland restoration

Authors: Keeley, J.E., et al.

Main findings (2-4 sentences): Prescribed burning in Stipa pulchra dominated California grasslands reduced non-native annual grass cover by 70% in the first post-fire year while having minimal impact on S. pulchra recovery. S. pulchra recovered to control levels within three years while non-native grasses remained suppressed until year five. Recovery patterns were duplicated after a second burn conducted five years later, demonstrating the reliability of the response.

Relevance to study (2-4 sentences): This paper directly supports prescribed burn as a restoration tool in Stipa pulchra dominated California grasslands using the same focal species and ecosystem as our study. The multi-year recovery pattern contextualizes the post-burn decline in our native grass cover and reinforces that our 2021-2025 window may not capture full recovery. We note that their study used repeated burns while ours analyzes a single burn event.

References

Bartolome et al. (2004) Hankins (2013) Berleman et al. (2016) Rau et al. (2008) Carlsen et al. (2017) Matthews et al. (2009) Keeley et al. (2023) Galloway et al. (2026) Bartolome et al. (2019) Menke (1992) Rickard (2024) Nestor (2025)

Bartolome, J. W., J. S. Fehmi, R. D. Jackson, and B. Allen-Diaz. 2004. Response of a native perennial grass stand to disturbance in california's coast range grassland. Restoration Ecology 12:279–289.

Bartolome, J., A. Brown, P. Hopkinson, M. Hammond, L. Macaulay, and F. Ratcliff. 2019. Evaluating prescribed fire effect on medusa head and other invasive plants in coastal prairie at point pinole. Grasslands 29.

Berleman, S. A., K. N. Suding, D. L. Fry, J. W. Bartolome, and S. L. Stephens. 2016. Prescribed fire effects on population dynamics of an annual grassland. Rangeland Ecology & Management 69:423–429.

Carlsen, T. M., E. K. Espeland, L. E. Paterson, and D. H. MacQueen. 2017. Optimal prescribed burn frequency to manage foundation california perennial grass species and enhance native flora. Biodiversity and Conservation 26:2627–2656.

Galloway, E. M., M. R. Musto, G. L. Vasey, and K. D. Holl. 2026. PRESCRIBED BURN IMPACTS ON CALIFORNIA COASTAL PRAIRIE SEED BANKS. Madroño 73.

Hankins, D. L. 2013. The effects of indigenous prescribed fire on riparian vegetation in central california. Ecological Processes 2:24.

Keeley, J. E., R. C. Klinger, T. J. Brennan, D. M. Lawson, J. La Grange, and K. N. Berg. 2023. A decade-long study of repeated prescription burning in california native grassland restoration. Restoration Ecology 31:e13939.

Matthews, J. W., G. Spyreas, and A. G. Endress. 2009. Trajectories of vegetation-based indicators used to assess wetland restoration progress. Ecological Applications 19:2093–2107.

Menke, J. W. 1992. Grazing and fire management for native perennial grass restoration in california grasslands. Fremontia 20:22–25.

Nestor, A. 2025. Evaluating sheep grazing as a strategy for restoring the north campus open space (NCOS) grasslands.

Rau, B. M., J. C. Chambers, R. R. Blank, and D. W. Johnson. 2008. Prescribed fire, soil, and plants: Burn effects and interactions in the central great basin. Rangeland Ecology & Management 61:169–181.

Rickard, A. 2024. North campus OpenNorth campus open space restoration project monitoring report: Year 7, december 2024.